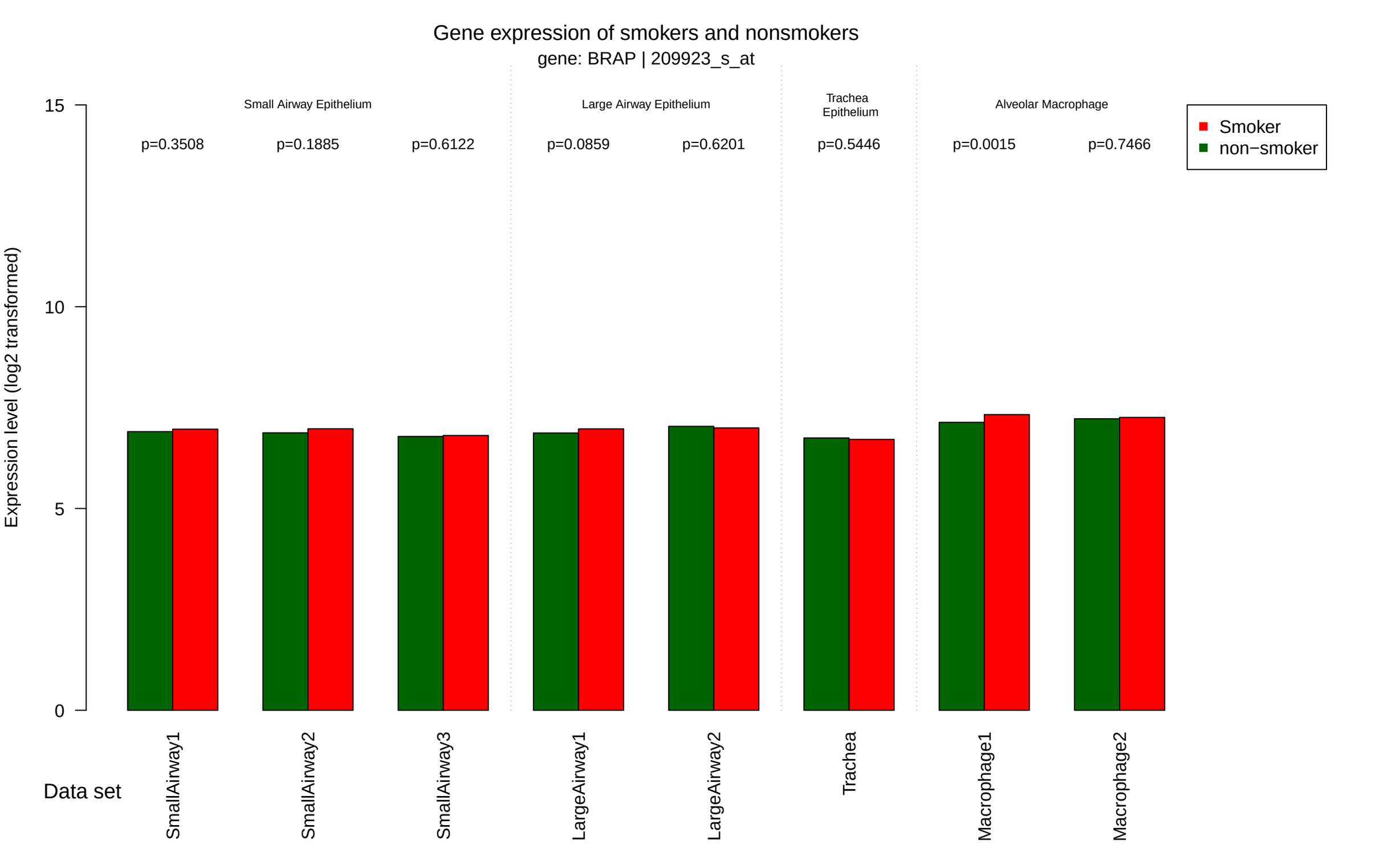
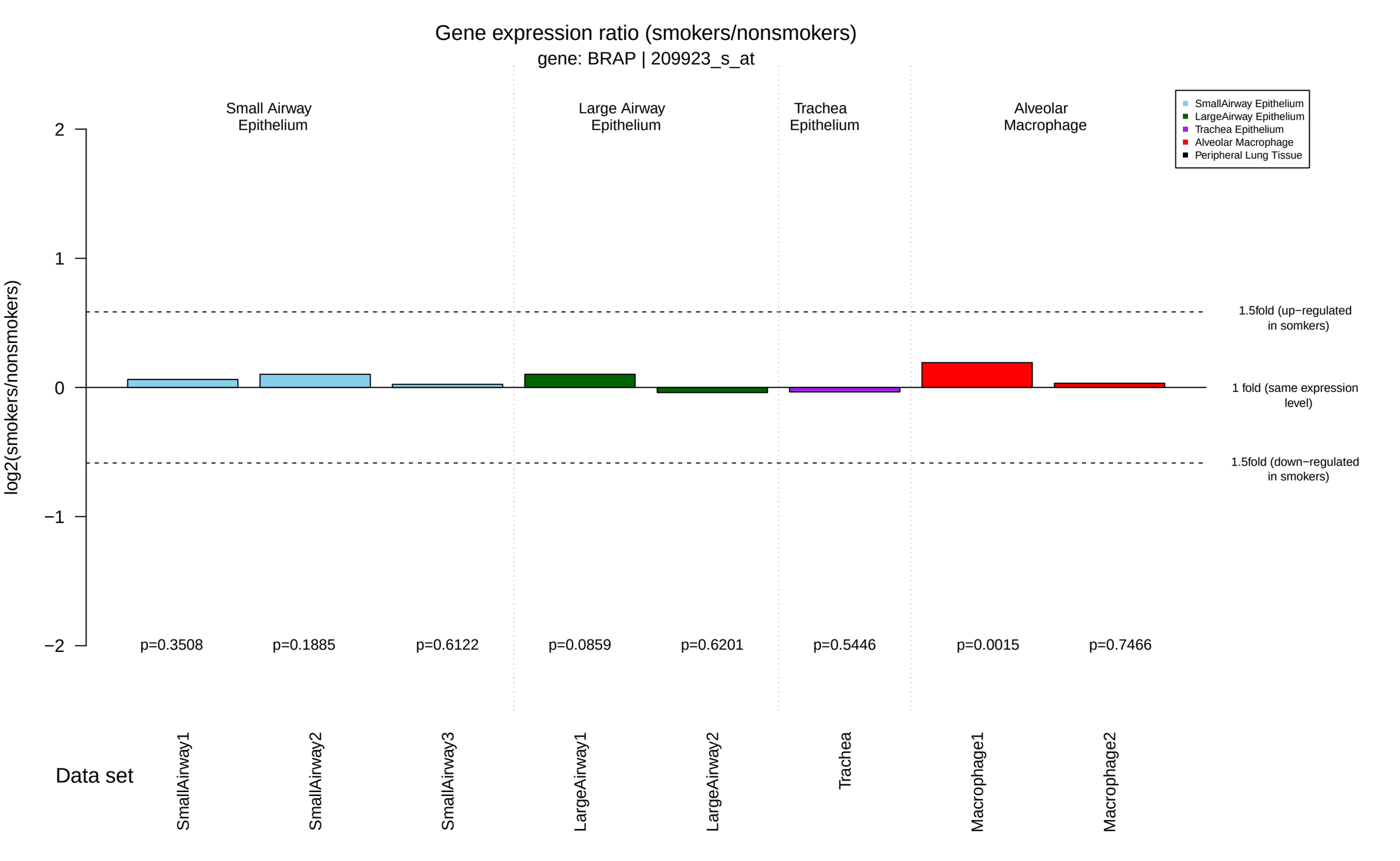
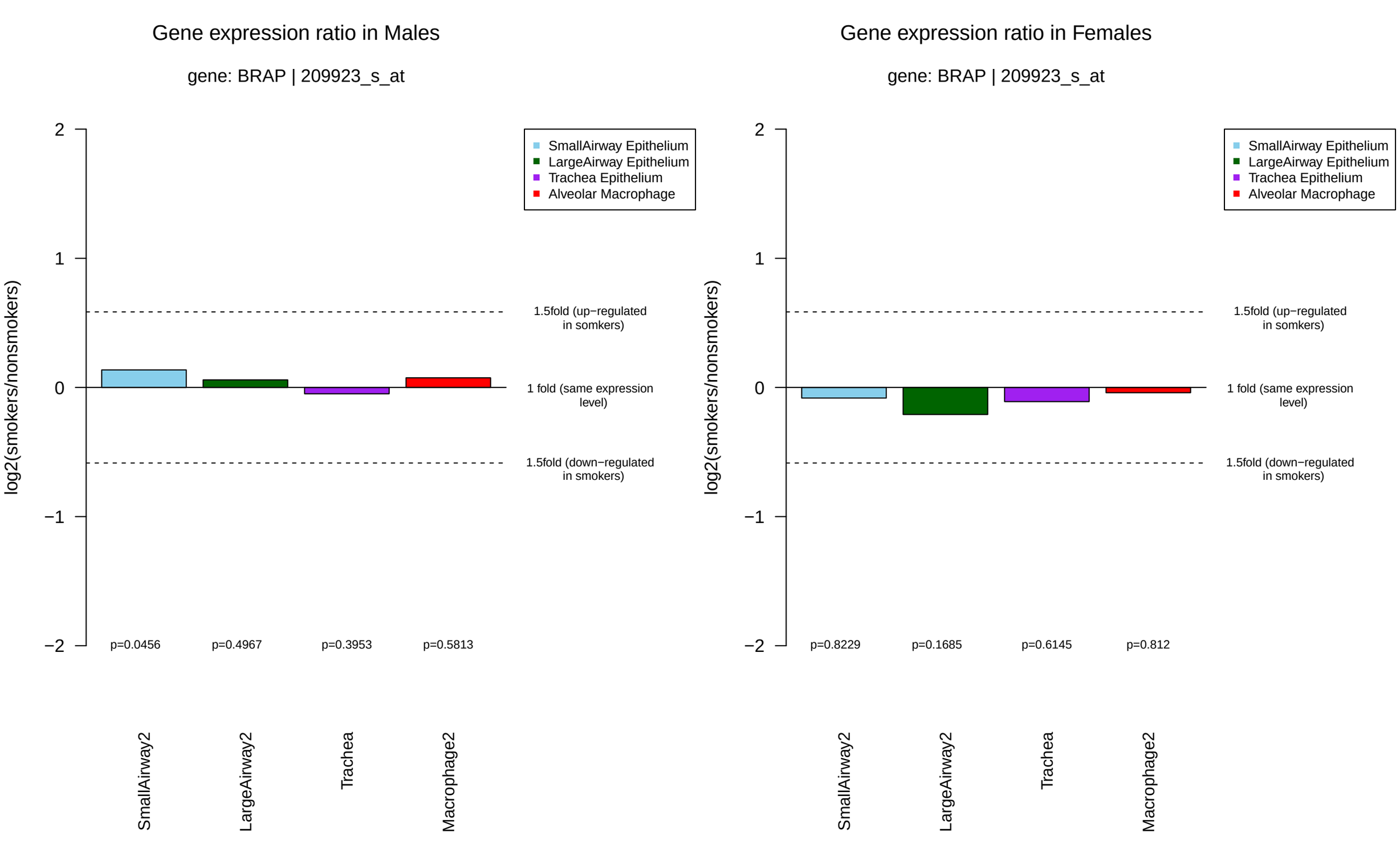
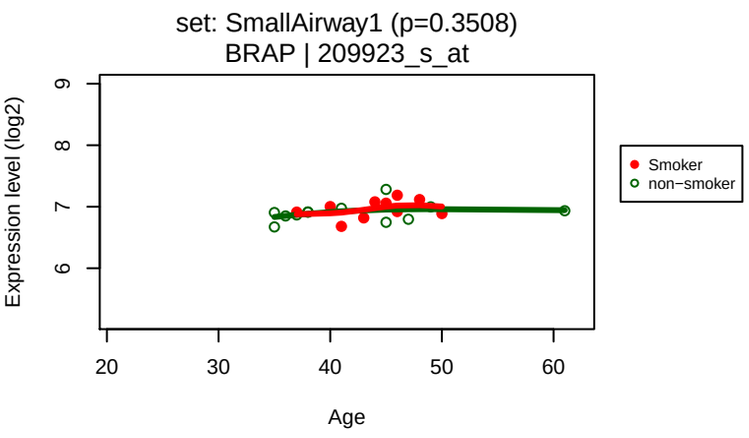
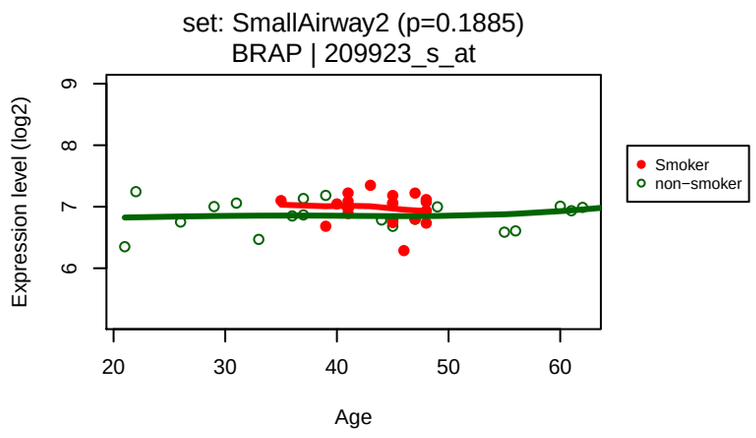
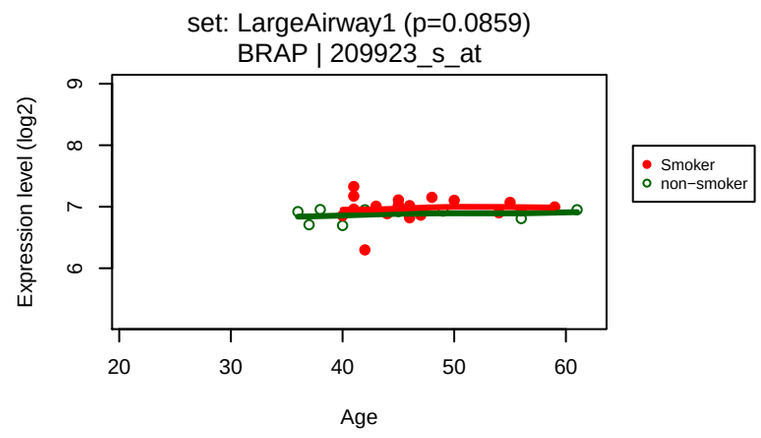

Gene expression of smokers and nonsmokers
gene: BRAP | 209923_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3508
p=0.1885
p=0.6122
p=0.0859
p=0.6201
p=0.5446
p=0.0015
p=0.7466
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

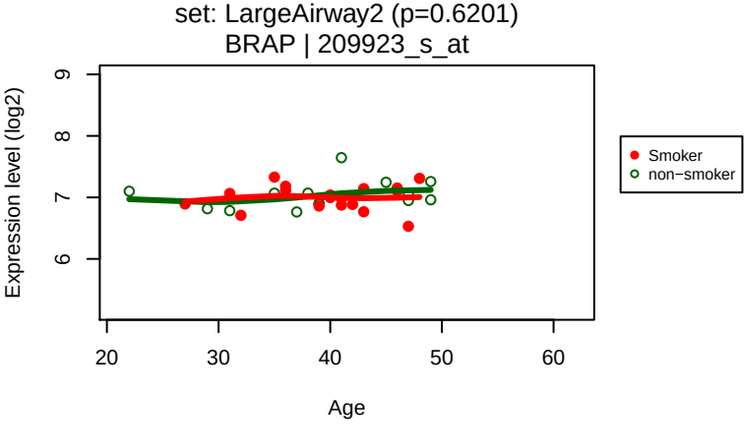

Gene expression ratio (smokers/nonsmokers)
gene: BRAP | 209923_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3508
p=0.1885
p=0.6122
p=0.0859
p=0.6201
p=0.5446
p=0.0015
p=0.7466
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

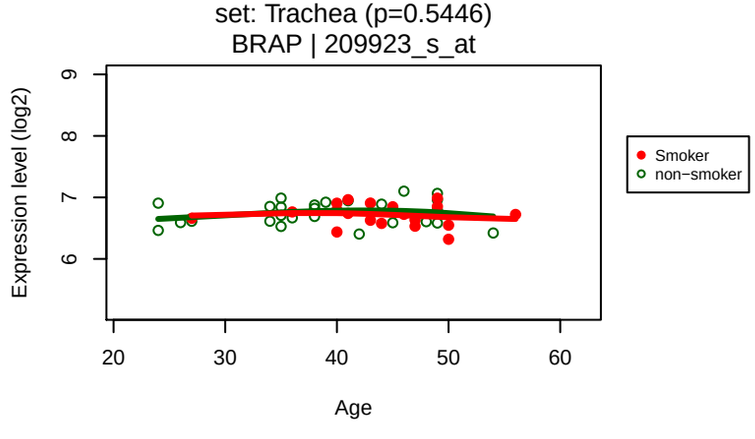

Gene expression ratio in Males
gene: BRAP | 209923_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0456
p=0.4967
p=0.3953
p=0.5813
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRAP | 209923_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8229
p=0.1685
p=0.6145
p=0.812
SmallAirway2
LargeAirway2
Trachea
Macrophage2

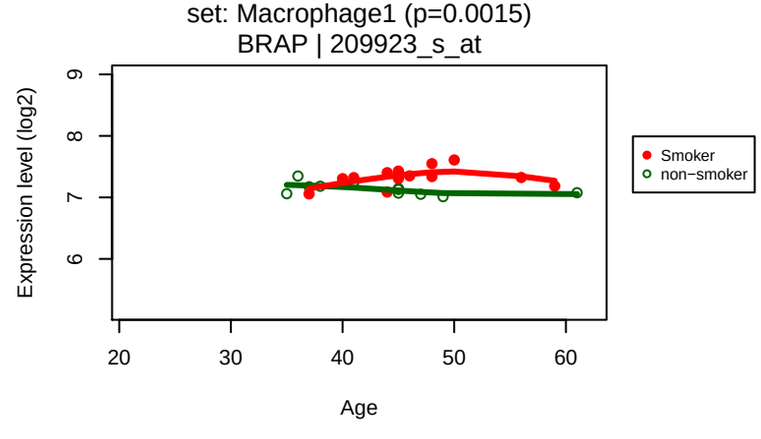

set: SmallAirway1 (p=0.3508)
BRAP | 209923_s_at
Expression level (log2)
Age
Smoker
non−smoker

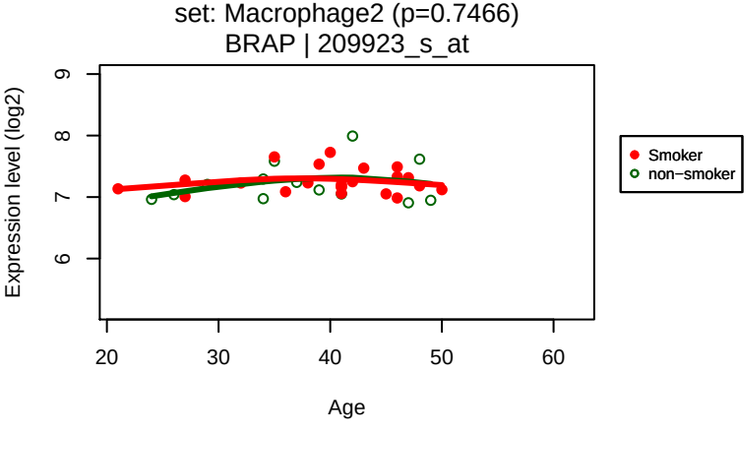

set: SmallAirway2 (p=0.1885)
BRAP | 209923_s_at
Expression level (log2)
Age
Smoker
non−smoker

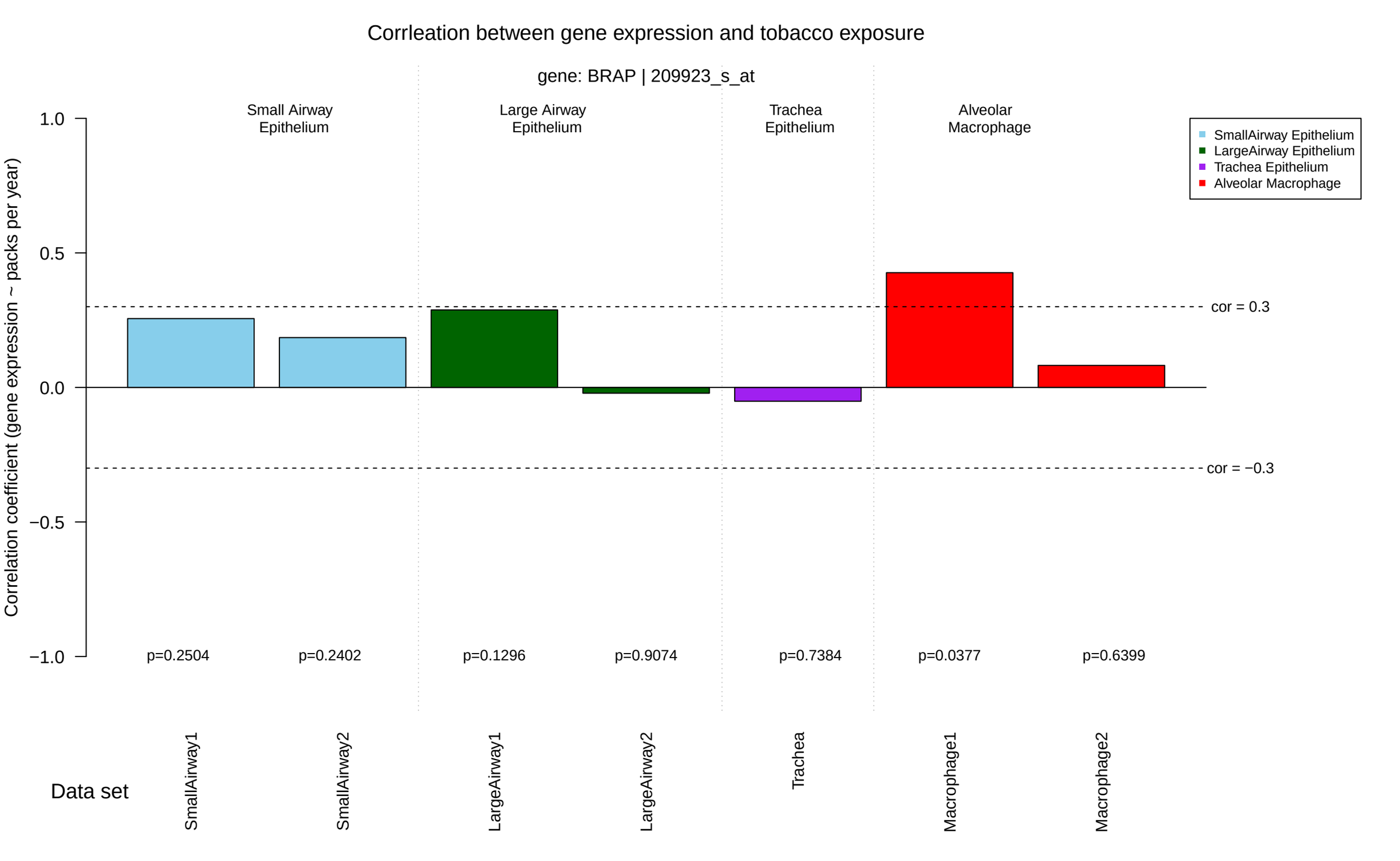

set: LargeAirway1 (p=0.0859)
BRAP | 209923_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6201)
BRAP | 209923_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5446)
BRAP | 209923_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0015)
BRAP | 209923_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7466)
BRAP | 209923_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRAP | 209923_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2504
p=0.2402
p=0.1296
p=0.9074
p=0.7384
p=0.0377
p=0.6399
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2